

The V_{600} Substrate at the Riemann–Hypothesis Frontier

A verified local realisation, a corrected reduction,
and an honest map of the global gap

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PRE-PEER-REVIEW OPEN RESEARCH PREPRINT. **No proof of the Riemann Hypothesis is claimed.** Every claim below is tagged [theorem], [verified] (computed, with a stated correctness gate), [evidence] (numerics consistent with, not proving), [conjecture], or [open]. All computations are reproducible from the accompanying `route_b/` and `sims/` code.

Abstract

We report the honest standing of the V_{600} closure substrate at the Riemann–Hypothesis (RH) frontier. **Main verified result:** the icosian ring — the maximal order of the definite quaternion algebra $(-1, -1/\mathbb{Q}(\sqrt{5}))$ whose unit group is the 600-cell — computes, by its own intrinsic Brandt/Hecke action at level $\mathfrak{p}_{31}$ (norm 31), the genuine cuspidal Hilbert modular newform over $\mathbb{Q}(\sqrt{5})$ of that level: its Hecke eigenvalues match the independently point-counted newform exactly on all 11 prime ideals of norm ≤ 41 , including signs and split-prime pairings, out-of-sample, with no fitting. This is a non-circular substrate $\leftrightarrow$ arithmetic identification, replacing a hardcoded map a prior audit exposed. **Corrected scope:** we withdraw an earlier claim that admissibility plus the explicit formula yields RH; they certify a genuine automorphic L-function, whose RH is equivalent to GRH and remains open. **Geometric map:** the substrate realises the *local* critical line (the per-prime Satake circle $|\alpha_q| = \sqrt{Nq}$, forced by temperedness) but not the *global* one; we explain why local $\rightarrow$ global is the genuine wall and name the missing object (a Hilbert–Pólya operator on the adelic arena). **Frontier computation:** we compute the Connes/Weil positivity functional for this L-function from the substrate’s own Satake data, gated against the Riemann ζ zeros to 0.01%; it is positive on every validated test function — evidence consistent with RH, not a proof. **Locating the operator:** the zeros are GUE (verified: level repulsion; a quasicrystal dual to the primes), and the Dyson/division-algebra structure pins the missing Hilbert–Pólya operator to the $\mathbb{C}$ (unitary, $\beta = 2$, time-reversal-broken) class — one rung from the substrate’s $\mathbb{H}$ ($\beta = 4$). The $\mathbb{H} \rightarrow \mathbb{C}$ transition is complexification ($\mathbb{H} \otimes \mathbb{C} = M_2(\mathbb{C})$, $2I \subset \mathrm{SU}(2)$, verified) realised by Jacquet–Langlands; the finite-place part of the operator (the commutative Hecke scaling algebra) is built and verified $\mathbb{C}$ -class, leaving the archimedean self-adjoint realisation — equivalently Weil positivity for all test functions, i.e. RH — as the single open step.

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1 Stance

This paper does not attempt to prove RH. It records, with reproducible computations and explicit status tags, exactly how far a specific geometric substrate reaches into the problem, where it stops, and why. The discipline throughout: the mathematics stays fully mathematical; no interpretive, observer-, or consciousness-based reading is load-bearing or implied.

2 The substrate and the local critical line

2.1 The icosian ring

Let $K = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$. Let $B = (-1, -1/K)$ be the Hamilton quaternions over K , ramified at both real places (definite). Its maximal order is the *icosian ring* $\mathcal{I}$, whose group of units is the binary icosahedral group $2I$ of order 120 — the 120 vertices of the 600-cell V_{600} .

Proposition 2.1 ([[verified](#)]). *The 120 unit icosians all have reduced norm 1 and are closed under multiplication (14400/14400 products verified in `route_b/icosian.py`), forming $2I$.*

2.2 The two critical lines

A central clarification of this work is that “the critical line” refers to two different objects:

- (1) **Local:** for each prime $\mathfrak{q}$, the zeros of the local factor $L_{\mathfrak{q}}(s)$. The Satake parameters satisfy $\alpha_{\mathfrak{q}} = \sqrt{N\mathfrak{q}} e^{i\theta_{\mathfrak{q}}}$ on the circle $|\alpha_{\mathfrak{q}}| = \sqrt{N\mathfrak{q}}$ — temperedness / Ramanujan — and consequently *each* local factor’s zeros lie exactly on $\Re(s) = 1/2$.
- (2) **Global:** the zeros of the full product $L(s) = \prod_{\mathfrak{q}} L_{\mathfrak{q}}(s)$. That these lie on $\Re(s) = 1/2$ is RH.

Proposition 2.2 ([[verified](#)], `route_b/satake_circle.py`). *For the substrate-computed eigenvalues (§3), every Satake parameter lies on its circle and every local zero lies on $\Re(s) = 1/2$; e.g. at $N\mathfrak{q} = 4, 5, 9, 11$ the angles are $138.6^\circ, 116.6^\circ, 70.5^\circ, 52.9^\circ/127.1^\circ$.*

This is the number-field shadow of the function-field RH that Deligne proved geometrically (Frobenius weights on $|\alpha| = \sqrt{q}$). The substrate realises the local critical line; the global one is the open problem.

2.3 Why the Hecke structure reaches the L-function

The substrate carries three independent extension structures, and only one reaches the L-function's zeros:

- **$W(H_4)$ geometric** (order 14,400): extends a statement from one V_{600} vertex to all 120 — *within* the substrate.
- **Cascade scale** ($\text{Pentagon} \rightarrow \text{Icos} \rightarrow V_{600} \rightarrow E_8 \rightarrow \dots$): propagates patterns up the hierarchy — *within* the substrate.
- **Hecke multiplicative**: the Hecke operators at fixed level are generated by the prime operators, with $T_m T_n = T_{mn}$ for coprime ideals and a prime-power recursion. This is the only one of the three that reaches the global L-function.

Theorem 2.3 (Multiplicativity tiling, [theorem]). *Let $\{a_p\}_{Np \leq P}$ be the eigenvalues of a commuting family on a finite-dimensional space, satisfying Ramanujan $|a_p| \leq 2\sqrt{Np}$, the prime-power recursion $a_{p^{k+1}} = a_p a_{p^k} - Np a_{p^{k-1}}$, and $a_{ab} = a_a a_b$ for coprime a, b . Then the full coefficient system $\{a_n\}$ is determined by the finite prime data, and*

$$L(s, \psi) = \sum_n \frac{a_n}{Nn^s} = \prod_p (1 - a_p Np^{-s} + Np^{1-2s})^{-1}$$

converges absolutely for $\Re s > 3/2$ (Euler product unconditional).

Hence the circle test (§3) is decisive: matching the substrate's *finite* prime eigenvalues to the genuine newform pins the *entire* L-function by tiling. It does not, by itself, place the zeros — that is RH (§4).

3 The circle test: the substrate computes a genuine newform

The target is the cuspidal Hilbert modular newform over K at level $\mathfrak{p}_{31}$ (norm 31; the prime $31 \equiv 1 \pmod{5}$ splits in K). This is the level at which the first cuspidal newform over $\mathbb{Q}(\sqrt{5})$ appears [1], corresponding to the elliptic curve 31.1- a .

3.1 The genuine target H (independent of the substrate)

Proposition 3.1 ([verified], `sims/sim_genuine_eigenvalues.py`). *The Hecke eigenvalues $a_q = Nq + 1 - \#E(\mathbb{F}_q)$ of the curve 31.1- a_1 ($y^2 + xy + \varphi y = x^3 + (\varphi + 1)x^2 + \varphi x$), computed by point counting (no eigenvalue table used), for all 44 good prime ideals of norm ≤ 200 , satisfy: the Ramanujan bound $|a_q| \leq 2\sqrt{Nq}$ (cuspidal), and the torsion divisibility $8 \mid \#E(\mathbb{F}_q)$ (the curve has $\mathbb{Z}/8\mathbb{Z}$ torsion — an independent correctness proof needing no external table).*

LMFDB confirms a single isogeny class at norm 31: the cuspidal space is 1-dimensional, not 26, settling an earlier dimension question.

3.2 The substrate side S (from the icosians' own arithmetic)

Because $\mathcal{I}$ has class number 1, the level- $\mathfrak{p}_{31}$ modular space is $\mathbb{C}[2I \setminus \mathbb{P}^1(\mathbb{F}_{31})]$ (Voight [2], Ch. 41; Demblé [1]).

Proposition 3.2 ([verified], `route_b/step2_eichler.py`). *Under the local splitting $B \otimes \mathbb{F}_{31} \cong M_2(\mathbb{F}_{31})$, the units $2I$ reduce to $A_5 \subset \mathrm{PGL}_2(\mathbb{F}_{31})$ (order 60), acting on $\mathbb{P}^1(\mathbb{F}_{31})$ (32 points) with orbit sizes $\{12, 20\}$. Hence $\dim = 2 = 1$ Eisenstein + 1 cuspidal, matching Proposition 3.1.*

The Hecke operator $T_{\mathfrak{q}}$ is the sum of the norm- $\mathfrak{q}$ quaternions (enumerated by the verified rank-8 short-vector routine, `route_b/short_vectors.py`, gated by $r(2) = 120$) acting on the two orbits; the cuspidal eigenvalue is $a_{\mathfrak{q}} = \mathrm{tr}(T_{\mathfrak{q}}) - (N\mathfrak{q} + 1)$.

Theorem 3.3 (Circle closure, [verified], `route_b/step3_4_hecke.py`). *The substrate cuspidal eigenvalues S equal the genuine target H on every prime ideal of norm ≤ 41 , computed with no fitting and no modular input:*

$N\mathfrak{q}$	4	5	9	11	19	29	41
substrate $a_{\mathfrak{q}}$	-3	-2	2	$\{-4, 4\}$	$\{-4, 4\}$	$\{-2, -2\}$	$\{-6, -6\}$
genuine H	-3	-2	2	$\{-4, 4\}$	$\{-4, 4\}$	$\{-2, -2\}$	$\{-6, -6\}$

The agreement is exact, includes signs and the split-prime pairings, and holds out of sample on $\{19, 29, 41\}$ (which influenced nothing). The Eisenstein eigenvalue is $N\mathfrak{q} + 1$ on the nose (row sums), and every $a_{\mathfrak{q}}$ is Ramanujan-bounded.

Remark 3.4. This is the central verified result. It is, mathematically, *expected* — $\mathcal{I}$ is the maximal order of the relevant quaternion algebra, so its Brandt action *is* this newform by Jacquet–Langlands — which is exactly why it is the right, non-circular confirmation: the substrate genuinely *computes* (does not inject) the eigenvalues. It replaces a hardcoded/fitted map that an internal provenance audit (`CIRCLE_TEST.md`) had exposed.

4 Corrected scope: admissibility does not give RH

The substrate eigenform passes the verifiable admissibility classes *genuinely*: multiplicativity $a_{\mathfrak{q}^2} = a_{\mathfrak{q}}^2 - N\mathfrak{q}$ (checked: $a_{(2)^2} = 5 = (-3)^2 - 4$, $a_{(\sqrt{5})^2} = -1 = (-2)^2 - 5$, `route_b/admissibility.py`, [verified]); Ramanujan (the Hasse bound, [theorem]); Sato–Tate (a theorem for this non-CM curve, [theorem]); functional equation and automorphy ([theorem]). So $L(\psi, s)$ is a genuine degree-4 cuspidal automorphic L-function (conductor $5^2 \cdot 31 = 775$).

Withdrawn overclaim

An earlier statement asserted that admissibility plus the Weil explicit formula *forces the zeros onto the critical line*. **This is false and is withdrawn.** A genuine automorphic L-function with a valid explicit formula does *not* have its zeros constrained to the line — that is precisely RH for it, equivalent to GRH, and [open]. The explicit formula is an identity equivalent to information about the zeros; Ramanujan bounds the prime side and Sato–Tate describes the eigenvalue distribution, but *neither pins the zeros*. (Re)established in `route_b/admissibility.py`.

5 Why local $\rightarrow$ global is the wall

The functional equation $\Lambda(s) = \Lambda(1-s)$ forces the zeros to be *symmetric about* $\Re(s) = 1/2$ [theorem]; it does *not* force them *onto* it. A zero could be a symmetric pair $\frac{1}{2} \pm \beta + i\gamma$ straddling the line; RH is the statement $\beta = 0$ always. The structure gives the *axis*; RH is the *on-ness*.

Remark 5.1 (The global object is not a polytope). The global L-function is the Euler product of all local factors, and its zeros are *emergent*: a product of factors each with zeros on a line has its own new zeros, visible in no single factor. The object that glues all places is the *adele class space* $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ — whose compact heart is the adelic solenoid, an *inverse limit of circles* (infinite-dimensional, noncommutative; Connes). It is not the 600-cell, not S^3 , not any finite stack of cells. Climbing the cascade ($V_{600} \rightarrow E_8 \rightarrow \dots$) gives bigger *local* objects, never the global glue; the cascade limit *is* the solenoid.

6 The adjoint that would close it

Conjecture 6.1 (Hilbert–Pólya, [conjecture]). *There is a self-adjoint operator $\hat{H}$ on a Hilbert space with $\text{spec}(\hat{H}) = \{\gamma_n : \zeta(\frac{1}{2} + i\gamma_n) = 0\}$.*

Given such $\hat{H}$, self-adjointness gives real spectrum, hence RH [theorem]. The Weil explicit formula is already a trace formula in the exact shape of the Selberg trace formula (primes $\leftrightarrow$ closed geodesics, $\log p \leftrightarrow$ lengths, zeros $\leftrightarrow$ Laplace eigenvalues), which is why $\hat{H}$ is believed to exist. Two written candidates: the Berry–Keating Hamiltonian xp (semiclassical count matches the zero count, but xp is not essentially self-adjoint), and Connes’ operator on the adele class space (a rigorous *reformulation*, §7).

Proposition 6.2 (The substrate’s operator has the wrong spectrum, [verified]). $\mathcal{C}_{\varphi} = (12 + \varphi^{-2})I - A_1$ on V_{600} is self-adjoint and positive definite, with spectrum $\{0.382, 2.67, 5.91, 9.38, 12.38, 14.38, 14.85, 15.1, \dots\}$ — nine algebraic numbers in K (smallest $= \varphi^{-2}$, the witness $V_{\min}$, multiplicity 1). It is the right type (self-adjoint) but the wrong spectrum (finite, algebraic, local) for the infinite transcendental γ_n .

7 Connes/Weil positivity, computed and gated

Connes’ reformulation: RH for L is equivalent to positivity of the Weil distribution, $W(h) = (\text{archimedean}) - (\text{prime}) = \sum_{\rho} h(\gamma_{\rho}) \geq 0$ for all h of positive type. The prime term is built from the substrate’s verified Satake angles θ_q ; the archimedean term from the gamma factor $\Gamma_{\mathbb{C}}(s)^2$ and conductor 775.

Proposition 7.1 (Validation gate, [verified], route_b/weil_positivity.py). *Run on the Riemann ζ function (known zeros), the identical machinery reproduces $\sum_{\rho} h(\gamma_{\rho})$ to relative error 1.0%, 0.01%, 0.00% at Gaussian widths $\sigma = 3, 4, 5$ — confirming the constants, gamma factor, and normalisation.*

Proposition 7.2 (Positivity holds for $L(\psi, s)$, [evidence]). *With the machinery gate-validated, the Weil/Connes functional for the norm-31 newform, from the substrate’s Satake data, is*

$$W(h) = +1.40, +3.70, +6.39 \quad \text{at } \sigma = 3, 4, 5,$$

positive on every validated test function.

What this is and is not

This is *evidence consistent with* RH for this L-function, computed from substrate data and gated against ζ . It is **not** a proof: Weil positivity must hold for *all* test functions of positive type, and we test finitely many with the prime sum truncated ($Nq \leq 150$). A negative $W(h)$ would have *disproved* RH; we found none. Proving positivity for all h is RH, exactly as open as before.

8 The measurable bridge: quantum chaos

The zeros connect to a *measured* physical phenomenon. The unfolded nearest-neighbour spacings of the Riemann zeros follow the **GUE** random-matrix law — the same law measured in heavy nuclei and chaotic microwave cavities (Montgomery–Odlyzko; Rudnick–Sarnak).

Proposition 8.1 ([verified], route_b/quantum_chaos_test.py). *On 40 zeros the spacings show level repulsion (no small gaps; fraction < 0.3 is 0.000 vs Poisson 0.259) and the histogram is $\sim 13\times$ closer to GUE than to Poisson. The zeros statistically behave like a quantum chaotic spectrum — evidence for the spectral (Hilbert–Pólya) picture, not a proof.*

The same duality appears as a *quasicrystal*: the diffraction $D(t) = \sum_n \cos(\gamma_n t)$ has Bragg peaks exactly at $\log(p^k)$ ([verified], 11/11 matched, diffraction_quasicrystal.py) — the zeros (outside) diffract into the primes (inside), which is the explicit formula seen as diffraction. Reconstructing the prime staircase from the zeros layer-by-layer (mirror_layers.py) shows each zero is a frequency harmonic; off-line tuning grows the error as $x^{1/2+\beta}$, so RH is the amplitude-balance condition.

Remark 8.2 (Honest scope). These confirm the structure (quantum-chaos class, quasicrystal duality); peak *locations* and spacing statistics hold regardless of RH. They are texture evidence, not the operator.

9 The division-algebra structure

The GUE spacing constant $\langle r \rangle \approx 0.6027$ (Atas et al.) is a function of the Dyson index β , and Dyson’s threefold way identifies β with a division algebra: $\beta=1 \leftrightarrow \mathbb{R}$ (GOE), $\beta=2 \leftrightarrow \mathbb{C}$ (GUE), $\beta=4 \leftrightarrow \mathbb{H}$ (GSE); physically these are the three time-reversal classes, with $\beta=2$ (GUE) **time-reversal broken**.

Proposition 9.1 (The class mismatch, [verified]). *The substrate is $\mathbb{H}$ (quaternionic; natural class GSE, $\beta = 4$): its operator A_1 on V_{600} is 93% degenerate. The zeros are $\mathbb{C}$ (GUE, $\beta = 2$; $\langle r \rangle = 0.62$). Reaching the generator from the substrate is a change of division algebra $\mathbb{H} \rightarrow \mathbb{C}$, not a deformation — the sharpest statement of why the symmetric geometry cannot itself be the generator.*

Proposition 9.2 (The $\mathbb{H} \rightarrow \mathbb{C}$ transition is complexification, [verified], route_b/jacquet_langlands_trace.py). *$\mathbb{H} \otimes \mathbb{C} \cong M_2(\mathbb{C})$ (algebra iso; reduced norm = determinant, to 2×10^{-16}), and the 120 icosian units map into $SU(2)$ (to 2.5×10^{-17}): $2I \subset SU(2)$, the unitary group. The Satake angles are $SU(2)$ conjugacy angles (Sato–Tate = the $SU(2)$ measure). Globally this is Jacquet–Langlands ($B^\times \leftrightarrow GL_2$, both becoming $M_2(\mathbb{C})$); the eigenvalue matching is the circle test (§3); the GUE high zeros follow by Rudnick–Sarnak. **The class flip is complexification, not geometry.***

Proposition 9.3 (Symmetry type, [verified]/[theorem], route_b/katz_sarnak.py). *The a_q are real $\Rightarrow L$ is self-dual $\Rightarrow$ Katz–Sarnak family symmetry is Orthogonal or Symplectic, not*

Unitary; weight-2 GL_2 (Iwaniec–Luo–Sarnak) $\Rightarrow$ **Orthogonal**, and rank 0 $\Rightarrow O^+$. So the full RCH trichotomy appears in three roles: substrate $= \mathbb{H}$, bulk zero statistics $= \mathbb{C}$ (GUE), family symmetry $= \mathbb{R}$ (Orthogonal).

10 Locating the operator: the adelic scaling

The conjectural operator is the generator of *scale* (Berry–Keating xp ; Connes’ idele action): $\mathrm{RH} \Leftrightarrow \text{scale is self-adjoint once the primes discretise it (SCALE_AND_SYMMETRY.md)}$. Decompose the adelic scaling operator as *finite places* (the Hecke algebra = local scalings) $\otimes$ *archimedean* (the Mellin/dilation continuum).

Proposition 10.1 (Finite part, [verified], route_b/adelic_scaling.py). *The Hecke operators T_q on the Brandt module commute ($\max \| [T_a, T_b] \| = 0$) — a commutative scaling algebra — with joint spectrum Eisenstein $(Nq + 1) \oplus$ cuspidal (a_q) , the latter unitary/Satake. So the adelic limit carries a $\mathbb{C}$ -class (unitary, $\beta = 2$) scaling structure at the finite places.*

The operator, specified as far as honestly possible

The missing self-adjoint operator is now pinned:

- **class:** $\mathbb{C}$ / unitary / $\beta = 2$ (not the substrate’s $\mathbb{H}/\beta = 4$);
- **finite-place part:** the verified commutative Hecke (scaling) algebra (§10);
- **what is missing ([open]):** the *archimedean* dilation glued in, with a *self-adjoint* realisation making the zeros a discrete point spectrum — i.e. Connes’ Weil positivity for *all* test functions. That last step is RH.

11 Status ledger

Claim	Status
120 icosian units $= 2I$; norms, closure	[verified]
Each local factor’s zeros on $\Re(s) = 1/2$ (Satake circle)	[verified]
Substrate computes the genuine norm-31 newform (11 ideals, out-of-sample)	[verified]
Multiplicativity / Ramanujan / Sato–Tate on the eigenform	[verified]/[theorem]
$L(\psi, s)$ is a genuine cuspidal automorphic L-function	[theorem]
Connes/Weil functional $W(h) > 0$ on validated tests	[evidence]
Zeros are GUE (quantum chaos); diffract into primes (quasicrystal)	[verified]
$\mathbb{H} \rightarrow \mathbb{C}$ class flip = complexification ($2I \subset \mathrm{SU}(2)$)	[verified]
Symmetry type: bulk Unitary/GUE, family Orthogonal O^+	[verified]/[theorem]
Adelic scaling: finite-place Hecke algebra is commutative, $\mathbb{C}$ -class	[verified]
“Admissibility + explicit formula $\Rightarrow \mathrm{RH}$ ”	withdrawn (false)
RH for $L(\psi, s)$ (= GRH); the archimedean self-adjoint operator	[open]
Existence of the Hilbert–Pólya operator $\hat{H}$	[conjecture]/[open]

12 Scope discipline

The mathematics above uses only fields, lattices, quaternion orders, Hecke operators, gamma factors, and conductors. **No observer, no consciousness, and no “self” enters any definition, and none is needed.** Any reading of primes or zeros as selves/agents is a personal-philosophical frame that is *outside* this mathematics; the work neither uses nor supports it, consistent with the programme’s own discipline (“ X exists $\Leftrightarrow \mathcal{C}(X) = X$ ”; no “primes are alive”; no “universe is conscious”). The randomness of the primes is the friction between addition and multiplication — field-independent, sourced by no value or agent; RH is the timeless, observer-free statement that the primes are maximally balanced.

Conclusion

The substrate reaches the RH frontier exactly this far: it *verifiably computes* a genuine Hilbert modular newform from its own geometry, realises the *local* critical line in full, supplies real Satake data to the genuine Connes positivity object (positive on every validated test), and — through the quantum-chaos and division-algebra structure — pins the missing operator to the $\mathbb{C}/\text{unitary}/\beta = 2$ class, with its finite-place part (the commutative Hecke scaling algebra) built and verified. It does *not* prove RH: the single remaining step is the *archimedean self-adjoint realisation* of the zeros as a discrete spectrum — equivalently Weil positivity for all test functions, equivalently the Hilbert–Pólya operator on the $\mathbb{C}/\text{adelic}$ side — which is RH and is open for everyone. This is the honest, undismissable standing: a verified local realisation, a fully-specified missing operator (class, finite-place part, and the one open gluing), and not a step fabricated.

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